Corporate Inventory Replenishment & Reorder Policy
Document Name: Corporate Inventory Replenishment & Reorder Policy
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SECTION 1: Scope, Purpose & Governance
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This policy governs all inventory management, stock level audits, and automated replenishment proposals across all regional fulfillment cente

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SECTION 2: Calculation of Available Inventory Position
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Available inventory position determines the actual stock available to satisfy demand and pipeline replenishment.
Rule 2.1: Available inventory position MUST strictly be computed using the deterministic equation:
    Available Inventory Position = On-Hand Stock - Reserved Stock + Confirmed Incoming Stock
Rule 2.2: Unconfirmed purchase orders or unverified incoming shipments SHALL NOT be counted towards available inventory under any circu
Rule 2.3: Reserved stock represents inventory allocated to confirmed customer sales orders awaiting shipment and must always be deducted

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SECTION 3: Safe Daily Sales Velocity & Data Sufficiency
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Rule 3.1: Average Daily Sales (ADS) velocity must be derived from historical sales transactions across a rolling lookback window (default: 30
Rule 3.2: INSUFFICIENT DATA GUARDRAIL: If a SKU has no historical transaction records in the specified lookback window, or has fewer th
Rule 3.3: In the case of insufficient data (such as product P900), the system must immediately flag status as "INSUFFICIENT_DATA", set rec

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SECTION 4: Reorder Point (ROP) Determination
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Rule 4.1: The Reorder Point triggers a replenishment action when Available Inventory Position falls strictly below this threshold.
Rule 4.2: The Reorder Point MUST be deterministically computed as:
    Reorder Point (ROP) = (Average Daily Sales * Supplier Lead Time Days) + Safety Stock
Rule 4.3: Arithmetic operations must be executed deterministically in Python code. An LLM or language model shall never be permitted to per
Rule 4.4: Safety Stock is defined as the buffer stock required to protect against demand volatility and lead-time delays, as designated in maste

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SECTION 5: Order Quantity Optimization & Lot Sizing
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Rule 5.1: If Available Inventory Position is greater than or equal to the Reorder Point, no order should be generated (Recommended Quantity
Rule 5.2: When Available Inventory Position falls strictly below the Reorder Point, the baseline shortage is:

Gross Shortage = Target Stock Level - Available Inventory Position

Rule 5.3: Minimum Order Quantity (MOQ) Constraint: The recommended order quantity must never be less than the Supplier Minimum Order

Rule 5.4: Pack Size / Lot Ceiling Rule: Order quantities must be rounded up to the nearest integer multiple of the Supplier Pack Size using cei

Final Recommended Quantity = ceil(max(Gross Shortage, MOQ) / Pack Size) * Pack Size

Rule 5.5: Every replenishment recommendation must articulate its assumptions, detailing the target stock level, available inventory position, a

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SECTION 6: Missing Data Handling, Interactive Prompts & Idempotency

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Rule 6.1: When essential data parameters?specifically Supplier Lead Time?are missing or null (e.g., product P204), the agent must not fabrica

Rule 6.2: Once the operator supplies the missing value, the agent resumes execution through the state machine.

Rule 6.3: Idempotency Key Registration: To eliminate duplicate order submissions during network retries or identical re-evaluations, every ger

Rule 6.4: If a duplicate plan request is processed, the system must recognize the existing idempotency key and return the original Plan ID and